Pavlo Radiuk

Associate Professor · Ph.D. in Computer Science
Department of Computer Science, Faculty of Information Technologies, Khmelnytskyi National University, Ukraine

radiukp@khmnu.edu.ua
radiukpavlo@gmail.com
radiukpavlo.github.io
Department profile
0000-0003-3609-112X
Google Scholar
Scopus 57216894492
Web of Science AAA-9727-2021
ResearchGate
LinkedIn
github.com/radiukpavlo
Khmelnytskyi, Ukraine

I work on explainable artificial intelligence: methods that map what a deep network has learned onto the features clinicians and engineers already use, so that its decisions can be checked and audited. I apply them in medical imaging, UAV inspection of energy infrastructure and urban traffic control, and I am a Researcher in the Horizon Europe project U_CAN.

Positions

2025 – present **Associate Professor**, Khmelnytskyi National University
2024 – present **Researcher, Horizon Europe project U_CAN**, Khmelnytskyi National University
2022 – present **Data Scientist (advisory)**, Horizon Dynamics
2022 – 2025 **Senior Lecturer**, Khmelnytskyi National University

Education

2025 – 2027 **D.Sc. (habilitation level)**, Information Technologies, Khmelnytskyi National University, in progress
2022 **Ph.D.**, Computer Science, Khmelnytskyi National University
2017 **M.Sc.**, Mathematical and Computer Modelling, Khmelnytskyi National University, with distinction

Research projects

2024 – 2028 **Ukraine towards Carbon Neutrality (U_CAN)**, Researcher, Khmelnytskyi traffic pilot
Horizon Europe, EU Mission for Climate-Neutral and Smart Cities, grant agreement 101148374



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2024 – 2026 **Intelligent System for Recognising Defects of Green-Energy Objects Using UAVs**, Researcher (executor)
Ministry of Education and Science of Ukraine, No. 0124U004665

Selected publications

76 works: 38 journal articles, 34 conference papers, 2 book chapters and 2 preprints. Google Scholar h-index 17 (1,158 citations) · Scopus h-index 10 · Web of Science h-index 7 (as of Oct 2026). Own name in bold; † marks a mentored junior researcher.

2026 **CARDIA-X: Global Semantic Transition and Rough-Set Rules for Auditable Post Hoc Electrocardiographic Explainability**
P. Radiuk, O. Barmak, L. Klymenko, I. Krak. *AI* 7(9):376. doi:10.3390/ai7090376

2026 **Equivariant Transition Matrices for Explainable Deep Learning: A Lie Group Linearization Approach**
P. Radiuk, O. Barmak, L. Bedratyuk, I. Krak. *Machine Learning and Knowledge Extraction* 8(4):92. doi:10.3390/make8040092

2025 **Criticality Assessment of Wind Turbine Defects via Multispectral UAV Fusion and Fuzzy Logic**
P. Radiuk, B. Rusyn, O. Melnychenko, T. Perzynski, A. Sachenko, S. Svystun†, O. Savenko. *Energies* 18(17):4523. doi:10.3390/en18174523

2025 **Hierarchical Deep Learning Model for Identifying Similar Targets in UAV Imagery**
D. Borovyk†, O. Barmak, **P. Radiuk**, I. Krak. *Drones* 9(11):743. doi:10.3390/drones9110743

2024 **Intelligent Integrated System for Fruit Detection Using Multi-UAV Imaging and Deep Learning**
O. Melnychenko, L. Scislo, O. Savenko, A. Sachenko, **P. Radiuk**. *Sensors* 24(6):1913. doi:10.3390/s24061913

2017 **Impact of Training Set Batch Size on the Performance of Convolutional Neural Networks for Diverse Datasets**
P. Radiuk. *Information Technology and Management Science* 20(1), pp. 20–24. doi:10.1515/itms-2017-0003

Service and recognition

PC member DEXA 2024–2026 · WISE 2024, 2026 · MobiWis 2026
Editorial Guest Editor, Sustainability (MDPI), 2024–2026; Volume Editor, CEUR Workshop Proceedings, Vol. 4141 (ExplAI-2025), 2025; Editorial Board Member and Handling Editor, Journal of Intelligent & Fuzzy Systems, 2023–2024.
Organising Organising Committee member, ExplAI workshop, 2025–2026; Lead organiser, U_CAN international conference on climate-neutral cities, 2026.
Peer review 450+ reviews for journals and conferences (Web of Science).
Awards Scholarship of the Cabinet of Ministers of Ukraine for Young Scientists, 2022–2024; Best Paper Award for Ukrainian authors, IDAACS-2023, 2023.
Teaching 8 current courses at bachelor's, master's and Ph.D. level, one taught in English.

Collaboration

Partners Open to Horizon Europe, MSCA and EuroHPC consortia as XAI methods lead, Medical AI partner, Energy & UAV inspection and Ukrainian pilot site (PIC 964860456). Partner kit: radiukpavlo.github.io/partner-kit.